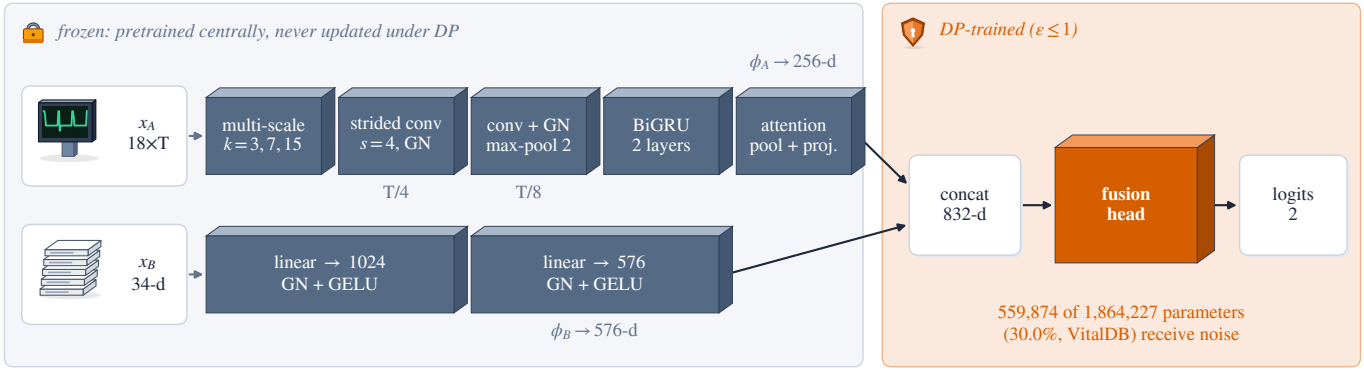
frozen: pretrained centrally, never updated under DP
$\phi_A \to 256$-d
x_A
18×T
multi-scale
$k = 3, 7, 15$
strided conv
$s = 4$, GN
conv + GN
max-pool 2
BiGRU
2 layers
attention
pool + proj.
T/4
T/8
x_B
34-d
linear → 1024
GN + GELU
linear → 576
GN + GELU
$\phi_B \to 576$-d
DP-trained ($\varepsilon \leq 1$)
concat
832-d
fusion
head
logits
2
559,874 of 1,864,227 parameters
(30.0%, VitalDB) receive noise